FedHARP: Towards Multi-Station HRRP Recognition via Federated Prototype Learning against Non-IID Data

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***Abstract*—In Radar Automatic Target Recognition (RATR), multi-station collaborative sensing based on High-Resolution Range Profile (HRRP) can enhance recognition performance but faces challenges including data privacy, limited communication bandwidth, and severe "client drift" caused by deeply heterogeneous (Non-IID) data. This paper proposes a Federated Heterogeneous Alignment method for Radar Prototypes (FedHARP). We design a Heterogeneous Prototype Alignment (HPA) mechanism, where each station shares only lightweight class prototypes, and the server generates global consensus prototypes via contrastive learning to guide local representation learning. Additionally, a Multi-factor Dynamic Weighting (MDW) strategy is introduced to dynamically optimize aggregation weights, mitigating performance degradation in heterogeneous environments. High-fidelity simulations based on measured data demonstrate that FedHARP significantly outperforms mainstream federated learning algorithms in recognition accuracy, approaching the upper bound of centralized training, thus providing a practical solution for distributed radar target recognition systems.**

***Index Terms*—Radar Target Recognition, High-Resolution Range Profile (HRRP), Federated Learning, Non-IID, Prototype Learning.**

I. Introduction

Multi-station collaborative sensing based on High-Resolution Range Profile (HRRP) is an effective way to improve the performance of Radar Automatic Target Recognition (RATR) [1–4]. However, traditional centralized fusion faces challenges of data privacy and communication overhead [5]. Federated Learning (FL), a "data-local, model-moves" paradigm, offers a solution [6]. Nevertheless, in practical deployments, data from various radar stations exhibit significant Non-Independent and Identically Distributed (Non-IID) characteristics (e.g., in viewing angles, SNR, and class distributions) [7]. This causes standard FL algorithms like FedAvg to suffer from a "client drift" bottleneck, leading to severe performance degradation and even "negative transfer" [8].Although some general FL optimization algorithms exist, they mostly overlook the unique properties of radar HRRP data [9].

To address these challenges, this paper introduces the Federated Heterogeneous Radar Prototype Alignment (FedHARP) framework, making three contributions. First, we systematically formulate the multi-station HRRP recognition problem under federated learning, articulating challenges of data privacy, communication efficiency, and deep heterogeneity, while analyzing shortcomings of prior art [10]. Second, FedHARP features a heterogeneous prototype alignment mechanism that generates a global consensus prototype for privacy-preserving knowledge fusion. This is complemented by a multi-factor dynamic weighting strategy that assesses client contributions to enhance system robustness and efficiency. Finally, extensive experiments on a high-fidelity simulator built from real-world data validate that FedHARP substantially outperforms state-of-the-art baselines, with its performance approaching the centralized learning upper bound, thus demonstrating its significant potential for real-world applications.

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II. RELATED WORK

A. HRRP-based Radar Target Recognition

High-Resolution Range Profile (HRRP) reflects a target's scattering center distribution [11, 12]. While CNNs and RNNs excel in capturing local and sequential patterns [13, 14], they typically rely on centralized data. This reliance limits their use in distributed radar networks where data privacy, communication costs, and link bandwidth are primary concerns [15].

B. Federated Learning for Non-IID Data

Federated Learning (FL) enables collaborative training without centralizing raw data, preserving local privacy [16]. However, FedAvg suffers from a "client drift" bottleneck under Non-IID conditions, potentially leading to "negative transfer" [17, 18]. While optimizations like FedProx and MOON mitigate this, they often overlook the multi-dimensional heterogeneity (e.g., SNR and azimuth sensitivity) inherent in radar HRRP data [19, 20].

C. Federated Prototype Learning

Prototype learning enhances discriminability by utilizing class centroids in the latent space [21]. FedProto adopts this by sharing lightweight prototypes instead of full models [22]. Yet, radar prototypes still exhibit distribution shifts due to diverse observation conditions [23]. Since simple averaging cannot reconcile these semantic shifts, FedHARP introduces an explicit heterogeneous refinement and alignment mechanism for robust knowledge fusion.

III. METHOD

A. Problem Formulation and Framework Overview

The dual-stream design of FedHARP is motivated by the physical properties of radar sensing. Heterogeneous Prototype Alignment (HPA) is designed to mitigate the feature representation shift caused by varying radar viewing angles and observation conditions. Meanwhile, Multi-factor Dynamic Weighting (MDW) addresses the reliability issue where clients with low signal-to-noise ratio (SNR) or divergent updates may negatively bias the global model.

Consider a federated learning system composed of K radar stations (clients) and a central server. The k-th radar station possesses a local private HRRP dataset $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$, where $x_i \in \mathbb{R}^d$ is an HRRP sample and $y_i \in \{1, \dots, C\}$ is its corresponding target class label. The data distribution across these stations exhibits Non-Independent and Identically Distributed (Non-IID) characteristics. Our objective is to collaboratively train an optimal global target recognition model $F(\theta) : \mathbb{R}^d \to \{1, \dots, C\}$, parameterized by θ, without sharing any raw data . The overall framework of the proposed system is illustrated in Fig. 1.

By incorporating the prototype knowledge stream into the classic model stream of federated learning, we introduce an enhanced framework referred to as Federated Learning with Hybrid Prototypes (FedHARP), which is elaborated in algorithm 1.

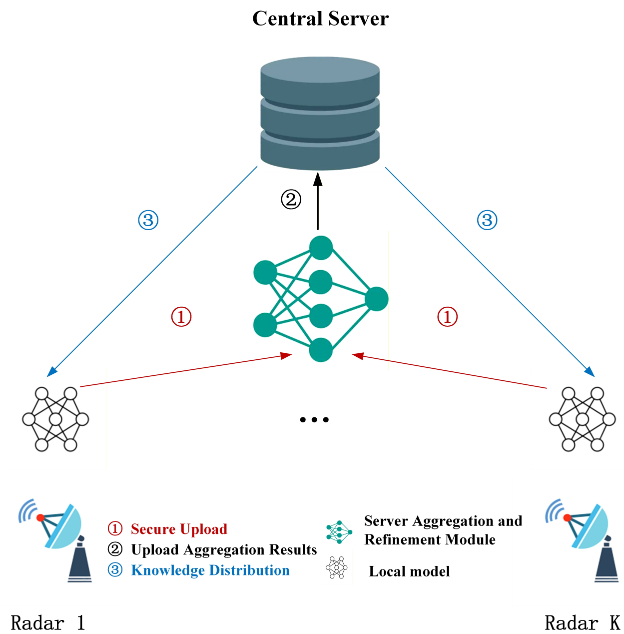

Fig. 1. Overall architecture diagram of the FedHARP framework.

Algorithm 1 FedHARP Framework

Require: K stations, T rounds, E epochs, learning rate η, initial parameters θ^0_{glob}, initial prototypes P^0_{glob}, local datasets $\{\mathcal{D}_k\}$.
Ensure: Final model θ^T_{glob}, global consensus prototypes P^T_{glob}.

for $t = 0$ to $T - 1$ **do**
  Phase 1 & 2: Distributed Local Training
  for each station $k \in \{1..K\}$ in parallel **do**
    Initialize $\theta^t_k \leftarrow \theta^t_{glob}$
    for $e = 1$ to E **do**
      $\theta^t_k \leftarrow \mathrm{SGD}(\mathcal{L}_{ce} + \lambda \cdot \mathcal{L}_{align})$
    end for
    Compute $P^t_k = \{p^c_k\}^C_{c=1}$ via f^e_k and **upload** (θ^t_k, P^t_k) to server
  end for
  Phase 3: Server Aggregation & Refinement
  Compute MDW weights a^t_k and aggregate: $\theta^{t+1}_{glob} = \sum^K_{k=1} a^t_k \theta^t_k$
  for each class $c = 1$ to C **do**
    Refine prototypes via g_ϕ and generate $\hat{p}^{t+1}_c = \frac{1}{K}\sum^K_{k=1} g_\phi(p^t_{k,c})$
  end for
  Update global prototype set: $P^{t+1}_{glob} = \{\hat{p}^{t+1}_c\}^C_{c=1}$
end for
return $(\theta^T_{glob}, P^T_{glob})$

B. Local Model for HRRP and Heterogeneous Prototype Alignment Mechanism

a) Local Model Architecture: The local model $f_k(\theta; x)$ at each radar station consists of a feature extractor $f^e_k(\theta_e; x)$ and a classifier $f^c_k(\theta_c; z)$, where $z = f^e_k(x)$ is the extracted feature representation. We design a lightweight network: it begins with a 4-layer 1D-CNN block to capture local range-cell patterns in the HRRP,followed by a Bidirectional Gated Recurrent Unit (Bi-GRU) layer with hidden dimension d_{hid} to model the contextual dependencies of the HRRP sequence, and finally, a channel attention module to enhance focus on discriminative scattering regions. The classifier is a fully connected layer.

b) Heterogeneous Prototype Alignment (HPA) Mechanism:

- **Local Prototype Calculation:** After each round of local training, for every class c, station k computes its local prototype, defined as the mean of the output features from the extractor for all samples belonging to that class:

$$p_k^c = \frac{1}{|\mathcal{D}_k^c|} \sum_{(x_i, y_i) \in \mathcal{D}_k^c} f_k^e(x_i) \tag{1}$$

where $\mathcal{D}_k^c$ is the set of samples at station k belonging to class c. The prototypes for all C classes constitute station k's local prototype set $P_k = \{p_k^1, p_k^2, \ldots, p_k^C\}$.

- **Server-Side Prototype Refinement:** The server collects the local prototype sets uploaded by all stations. For each class c, the server obtains a set of prototypes $\{p_1^c, p_2^c, \ldots, p_K^c\}$. Due to data heterogeneity, these same-class prototypes from different stations may be scattered in the feature space. We introduce a lightweight prototype refinement network $g_\phi(\cdot)$ (a two-layer MLP), which aims to map these prototypes into a new space through contrastive learning, pulling same-class prototypes closer together while pushing different-class prototypes apart.

Specifically, for a pair of prototypes (p_i^c, p_j^c) (same class, different stations), we treat it as a **positive pair**. For $(p_i^c, p_j^{c'})$ where $c \neq c'$, it is considered a **negative pair**. The training objective of the refinement network is to minimize the following contrastive loss:

$$\mathcal{L}_{\text{refine}} = -\frac{1}{|\mathcal{P}|} \sum_{(p_i, p_j) \in \mathcal{P}} \log \frac{\exp(z_{ij})}{\sum_{P_k \in \mathcal{N}_i} \exp(z_{ik})} \tag{2}$$

where $z_{ab} = \text{sim}\left(g_\phi(p_a), g_\phi(p_b)\right)/\tau$, $\mathcal{P}$ denotes the set of all positive sample pairs, $\mathcal{N}_i$ is the set of negative pairs for prototype p_i, $\text{sim}(\cdot,\cdot)$ is the cosine similarity function, and τ is a temperature parameter. After training, for each class c, its global consensus prototype $\hat{p}^c$ is obtained by averaging all refined prototypes belonging to that class:

$$\hat{p}^c = \frac{1}{K} \sum_{k=1}^{K} g_\phi\left(p_k^c\right) \tag{3}$$

The consensus prototypes for all classes form the global consensus prototype set $\hat{p} = \{\hat{p}^1, \hat{p}^2, \ldots, \hat{p}^C\}$.

- **Prototype-based Feature Alignment Training:** In the next round of local training, station k uses not only the standard cross-entropy classification loss $\mathcal{L}_{\text{ce}}$ but also introduces a prototype alignment loss $\mathcal{L}_{\text{align}}$. This alignment loss compels the local extracted features to move closer to the corresponding global consensus prototypes. It is defined as:

$$\mathcal{L}_{\text{align}} = \sum_{c=1}^{C} \sum_{x_i \in \mathcal{D}_k^c} \left\| f_k^e\left(x_i\right) - \hat{p}^c \right\|_2^2 \tag{4}$$

The total loss function for local training is:

$$\mathcal{L}_{\text{local}} = \mathcal{L}_{\text{ce}} + \lambda \cdot \mathcal{L}_{\text{align}} \tag{5}$$

where λ is a hyperparameter that controls the strength of the alignment. This loss function ensures that samples of the same class from different stations cluster around the consensus prototype in the feature space, thereby effectively aligning the heterogeneous data distributions.

C. Multi-factor Dynamic Weighting (MDW) Aggregation Strategy

The aggregation weight $w_k = n_k / \sum_{j=1}^{K} n_j$ in classic FedAvg is solely related to the data volume, which can cause the model to be biased towards clients with large but low-quality data (e.g., low SNR) in highly Non-IID scenarios. To address this, we propose a multi-factor dynamic weighting strategy. At aggregation round t, a dynamic weight $a_k^{(t)}$ is calculated for each client k:

$$a_k^{(t)} = \alpha \cdot s_{\text{qual}}^{(t)}(k) + \beta \cdot s_{\text{dir}}^{(t)}(k) + \gamma \cdot s_{\text{vol}}^{(t)}(k) \tag{6}$$

where α, β, $\gamma \geq 0$ and $\alpha + \beta + \gamma = 1$ are tunable hyperparameters. The three scoring terms are defined as follows:

- **Data Quality Score $s_{\textbf{qual}}^{(t)}(k)$:** Calculated based on the accuracy $\text{Acc}_k^{(t-1)}$ of client k's local model from the previous round on its local validation set, reflecting its data quality and model performance.

$$s_{\text{qual}}^{(t)}(k) = \frac{\exp\left(Acc_k^{(t-1)}/T_{\text{qual}}\right)}{\sum_{j=1}^{K} \exp\left(Acc_j^{(t-1)}/T_{\text{qual}}\right)} \tag{7}$$

- **Update Direction Score $s_{\textbf{dir}}^{(t)}(k)$:** Calculated based on the cosine similarity between client k's current model update $\Delta\theta_k^{(t)} = \theta_k^{(t)} - \theta^{(t)}$ and the average update direction $\overline{\Delta\theta}^{(t-1)}$ from the previous round, reflecting the consistency of its update with the global trend.

$$s_{\text{dir}}^{(t)}(k) = \frac{\exp\left(\text{cosine}\left(\Delta\theta_k^{(t)}, \overline{\Delta\theta}^{(t-1)}\right)/T_{\text{dir}}\right)}{\sum_{j=1}^{K} \exp\left(\text{cosine}\left(\Delta\theta_j^{(t)}, \overline{\Delta\theta}^{(t-1)}\right)/T_{\text{dir}}\right)} \tag{8}$$

- **Data Scale Score $s_{\textbf{vol}}^{(t)}(k)$:** Consistent with classic FedAvg, it is based on the proportion of local data volume.

$$s_{\text{vol}}^{(t)}(k) = \frac{n_k}{\sum_{j=1}^{K} n_j} \tag{9}$$

Finally, the global model is aggregated using the formula:

$$\theta^{(t+1)} = \sum_{j=1}^{K} a_k^{(t)} \theta_k^{(t)} \tag{10}$$

This strategy adaptively assigns higher weights to clients with high-quality data and update directions consistent with the global trend, thereby accelerating convergence and enhancing the performance of the global model.

IV. EXPERIMENTS AND RESULTS ANALYSIS

A. Experimental Setup

We used a measured HRRP dataset with 5 aircraft classes, simulating a 4-station heterogeneous environment via non-uniform azimuth partitioning, a label-wise Dirichlet distribution ($\alpha = 0.3$), and differentiated SNR (10–20 dB). The baselines include Local-Only, FedAvg, FedProx, FedProto, and Centralized. All methods used the Adam optimizer ($LR = 0.001$) for $T = 100$ rounds and $E = 5$ local epochs. All experiments were repeated 5 times. Although the experiment uses 4 stations and 5 classes, these are derived from high-fidelity real-world measured data across a wide range of aspect angles, providing a rigorous testbed for heterogeneous radar environments. Future work will extend to larger-scale datasets and hardware-in-the-loop validation.

B. Recognition Performance

As illustrated in Fig. 2, FedHARP demonstrates superior convergence speed and final accuracy compared to all federated learning baselines. It achieves 90% accuracy in merely 42 communication rounds, which is approximately 38% faster than FedAvg, highlighting its high communication efficiency. Notably, while FedProto also utilizes prototypes, FedHARP converges more stably and reaches a higher plateau, proving that our HPA refinement effectively mitigates the negative impact of local prototype shifts. The final test accuracy of FedHARP reaches 94.2%, closely approaching the performance upper bound of 96.5% set by centralized training.

C. Ablation Study

The ablation study results, as summarized in Table I, validate the contribution of each key component in FedHARP. Removing the Heterogeneous Prototype Alignment (HPA) mechanism leads to a performance drop of 4.6%, underscoring its critical role in facilitating cross-client knowledge transfer. Similarly, disabling the Multi-Dimensional Weighting (MDW) aggregation strategy results in a significant accuracy reduction of 2.9%, confirming the necessity of adaptive aggregation. The complete FedHARP framework achieves the best performance across all experimental settings.

D. Robustness under Data Heterogeneity

We further evaluate the model robustness under varying degrees of data heterogeneity, controlled by the Dirichlet parameter α. As shown in Fig. 3, all methods experience performance degradation as data heterogeneity increases (i.e., α decreases). Nevertheless, FedHARP maintains the strongest robustness. Under extreme heterogeneity ($\alpha = 0.1$), FedAvg even exhibits "negative transfer", performing worse than local training alone, while FedHARP retains a competitive accuracy of 80.1%. This resilience stems from the fact that our HPA mechanism maintains a consistent feature anchor via global prototypes, while the MDW strategy filters out divergent updates from stations with highly biased distributions, effectively preventing the global model from collapsing toward dominant but non-representative clients.

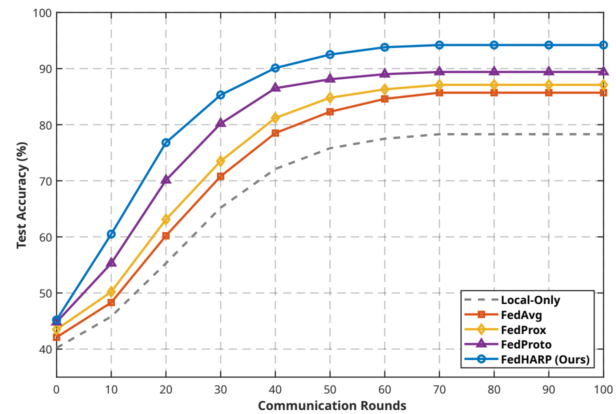

Fig. 2. Convergence Analysis.

TABLE I
ABLATION STUDY RESULTS

Model Variant	Accuracy (%)	Drop vs. Full (%)
FedHAPR (Full)	94.2	–
w/o HPA	89.6	4.6
w/o MDW	91.3	2.9
w/o HPA & MDW	87.1	7.1

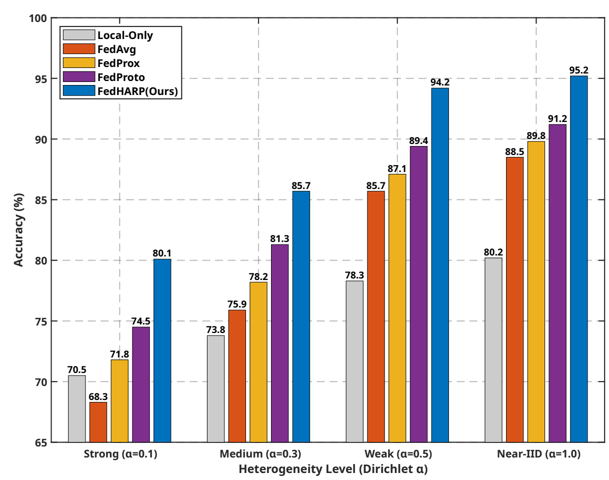

Fig. 3. Robustness analysis under varying data heterogeneity levels (Dirichlet α).

V. CONCLUSION

This paper proposes FedHARP, a federated learning framework that tackles data privacy and heterogeneity in distributed radar HRRP recognition. By integrating Heterogeneous Prototype Alignment (HPA) and Multi-factor Dynamic Weighting (MDW), FedHARP enables efficient, privacy-preserving knowledge fusion across radar stations. Experiments show it significantly outperforms mainstream methods in accuracy, convergence, and robustness, nearing centralized training performance. This work offers a practical path toward collaborative, secure distributed radar systems, with future efforts targeting category-heterogeneous extensions and real-world validation.

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